DS 116: What number goes in the brief?

Case study, single numeric variable (20 minutes)

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A development agency is writing a one-page brief on the world in 2007. The table has one row per country: life expectancy, GDP per capita (income, in international dollars), and population.

The data are Gapminder's 2007 country figures (Gapminder Foundation, CC BY 4.0). Stay with one variable at a time. Do not draw a scatterplot.

You have 20 minutes. Each task is one or two plots and one sentence.

```
library(ggplot2)
gap <- read.csv('Data/gapminder_2007.csv')
```

Task 1. What typical income should the brief quote? (7 minutes)

Income is `gdpPercap`.

1. Draw a histogram.
2. Draw the same histogram on a log10 scale (`scale_x_log10`).
3. Draw a normal Q-Q plot of `log10(gdpPercap)`.
4. Compute the mean, the median, and the geometric mean, `10 ^ mean(log10(gdpPercap))`.

Sentence: which of the three numbers belongs in the brief, and is income roughly log-normal?

Task 2. Is life expectancy the same kind of variable? (5 minutes)

Life expectancy is `lifeExp`.

Draw an eCDF and mark the median in red and the mean in green. Read which line sits further left.

Sentence: which way is the skew, and would a log scale be the right next step?

Task 3. Does the continent change the income number? (6 minutes)

1. Draw a boxplot of `gdpPercap` by `continent`, with the mean as a red point.
2. Draw the same boxplot with `scale_y_log10()`.

Oceania is Australia and New Zealand only.

Sentence: for which continent is the mean a poor summary of a typical country, and what do you do with Oceania?

Task 4 Check for lognormality

`pop` is the population of each country. Check whether a log makes it close to normal, and compute the mean, the median, and the geometric mean. Which number describes a typical country?